Loops # task 5

For Loop

1. Write a Python program to print numbers from **1 to 10** using a for loop.
2. Print numbers from **10 to 1** using a for loop.
3. Print all **even numbers from 1 to 50**.
4. Print all **odd numbers from 1 to 50**.
5. Print the **multiplication table** of a number entered by the user.
6. Find the **sum of numbers from 1 to 100**.
7. Find the **sum of all even numbers from 1 to 50**.
8. Find the **sum of all odd numbers from 1 to 50**.
9. Print the **square of numbers from 1 to 10**.
10. Print the **cube of numbers from 1 to 10**.
11. Ask the user for a number and find its **factorial** using a for loop.
12. Count the number of **digits** in a given number using a loop.
13. Find the **sum of digits** of a given number.
14. Print all **factors** of a given number.
15. Check whether a given number is **prime** using a for loop.
16. Print all **prime numbers between 1 and 100**.
17. Find the **largest number** in a list using a for loop.
18. Find the **smallest number** in a list using a for loop.
19. Count how many **even and odd numbers** are present in a list.
20. Count the number of **positive, negative and zero values** in a list.

- **While Loop**

21. Print numbers from **1 to 10** using a while loop.
22. Print numbers from **10 to 1** using a while loop.
23. Print all **even numbers from 1 to 50** using while.

24. Print all **odd numbers from 1 to 50** using while.

25. Find the **sum of numbers from 1 to N** using while.

26. Find the **factorial of a number** using while.

27. Reverse a given number using a while loop.

28. Find the **sum of digits** of a number using while.

29. Count the **number of digits** in a number using while.

30. Check whether a number is a **palindrome** using a while loop.

🔹 **Nested Loop / Pattern**

31. Print this pattern:

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*
**
***
****
*****
```

32. Print this pattern:

```
*****
****
***
**
*
```

33. Print this pattern:

```
1
12
123
1234
12345
```

34. Print this pattern:

1

22

333

4444

55555

35. Print this pattern:

*****

*****

*****

*****

*****

36. Print a **multiplication table from 1 to 10** using nested loops.

37. Print this pattern:

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

38. Print a **right-angled number triangle** using nested loops.

39. Print all combinations of two numbers where **both numbers range from 1 to 5**.

40. Write a program using nested loops to print:

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